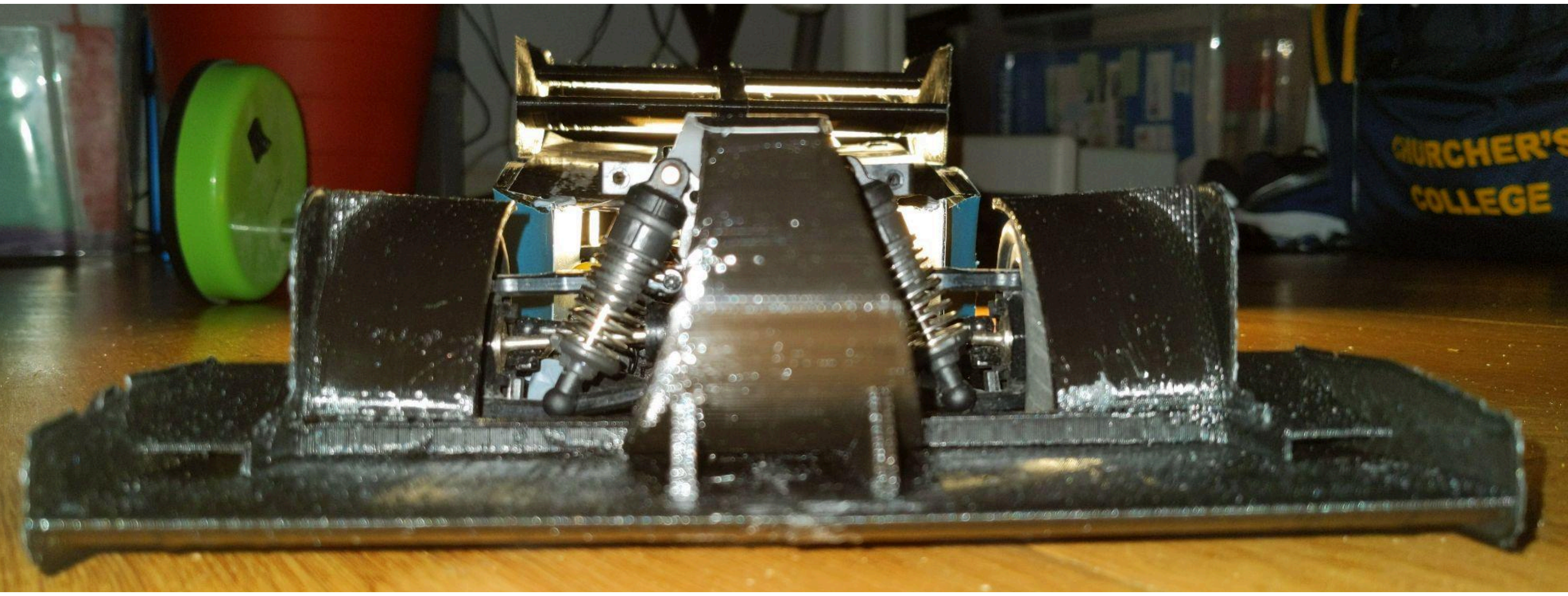


AERORC

Contents:

1.0 **Background**

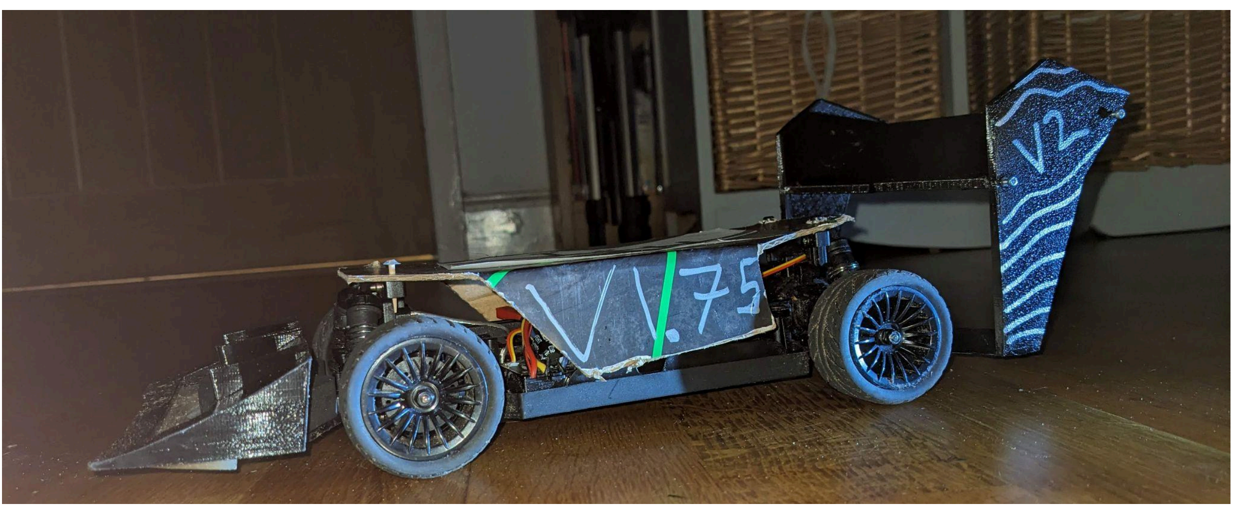
AERORC is essentially an aerodynamics project on wheels. After the last year I have slowly built up my knowledge on both aerodynamics and 3D Modelling, allowing me to build the final form of the project which I have called UDIPrototype. AERORC hasn't got the traditional motivation of a typical motorsports project. This car isn't built to win something, or take a lap record. Instead, the defining requirement is that it was cheap, and I have certainly succeeded in that. This led to a harsh compromise in choosing the UDIRC 1607 Brushed as the chassis and powertrain for my aerodynamics project. It's not particularly fast (I have been going on the assumption that it does 25kph, about half its rated top speed) and not particularly well built either. In fact, just undoing the screws in the rear of the chassis over and over has been enough to strip them twice. Over the course of the project I have gained 3 important and necessary skills to design and manufacture the high performance RC car of my future. 1.) Computational Fluid Dynamics and aerodynamics, 2.) conceptualising, designing and manufacturing parts for an RC car, and 3.) Making those parts as lightweight as possible within their strength requirements.

2.0 **AERORC V1 and 1.5**

V1 was shocking. It looks like an overfed batmobile. The mounting of the bodywork and its design was poor. More crucially though, its aerodynamic devices were most likely useless. I highly doubt this car generates any considerable downforce, even at top speed. AERORC V1's (and 2's) rear wing weighed a slightly ridiculous 65 grams, and wobbled a little whilst driving too. The body was a polystyrene sheet. After crashing the car (seen left) I replaced the entire rear wing which was the same in form but slightly different in mounting. I also gave it a card shell and dubbed the car V1.5.

These cars were nothing more than an introduction to myself of what was to come. Despite its looks though, it's what got me going on this whole project, and the mounting solution for the rear (although refined) has wiggled its way into every one of my cars.

3.0 **V2**

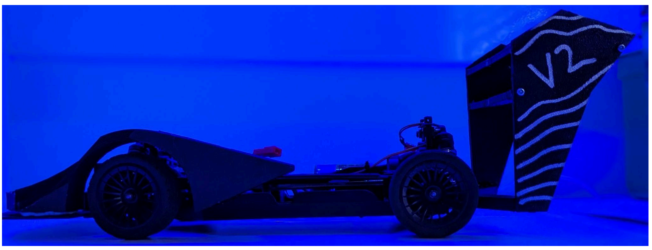


3.1 **V1.75:**

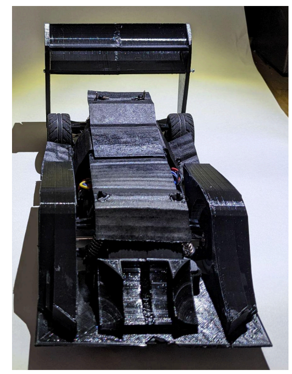
V2 Had a notable precursor with V1.75. I knew the first thing I wanted to do was change the rear wing, so I designed that before building and testing it with the old spec front and body. You can see above this 1.75 car. Note I had already painted the rear wing with V2, since I knew I would only test this car once or twice before completely overhauling the rest of the car.

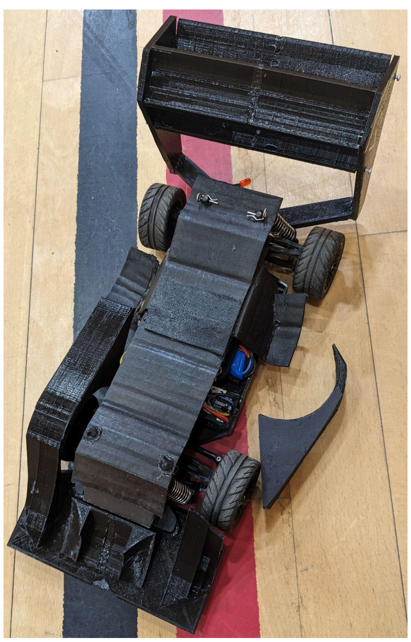
3.2 **V2 Rear Wing:**

Brief tangent over, the design of the V2 rear wing. I took advice from many in that the old rear wing was unnecessarily tall and thin. I did have reason behind this however, since I thought that printing anything larger would result in a drastic increase in weight. How wrong that was! The assembly decreased in weight (not by much but a little) and the surface area of the front (of the main elements) increased by ~3100mm to 9404mm. In retrospect an increase in frontal area is probably the most important thing in creating maximum lift in this scale and speed. Drawing knowledge from the rest of the rc world, it's fairly clear to see that wings are extremely rare. The more traditional approach is of course to add a large gurney flap to the rear. Replacing this traditional approach with my "real" wings is of course much harder to manufacture, but I am sure provides more performance.



3.3 **V2:** V2 was definitely a **huge** step forward in every respect. It set up the design philosophy for every car going forward. The wheel covers on the front solidified the concept that this car was not going to be open wheel. The splitter would stay for the next few iterations as well. Nonetheless this was a car plagued with design issues. The cardboard roof was relatively useless, getting wet and falling apart after only a few test runs. The gap between the top of the wheel cover and the wheel itself was way too large and the mount I used to connect the whole front end of the bodywork to the chassis used the old foam bumper mount. This led to a lot of issues with the mounting becoming looser during a test session. You will notice as well that a second element to the rear wing was added. Although performance wise I am certain this was fairly useless, it did help with the rear wing construction though. Essentially I had tried to heat screws up and melt/thread them through the side plates and into the wing. To be honest this didn't really work. I am sure with my current knowledge I could make this work, but simply glueing the element to the plate is **way** more effective. As well as this the technique of plastic welding both the two halves of the wing, and other parts of the car, has been crucial to the construction of most iterations of this car going forward. So in summary this car was **the** most crucial step forward in the project. Not in the performance of the car really, but in its construction and design. Its philosophy is the same philosophy I used with the UDIPrototype. All of the details I thought back then but didn't have the skills to produce, I have produced.

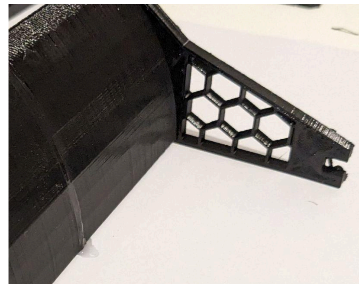


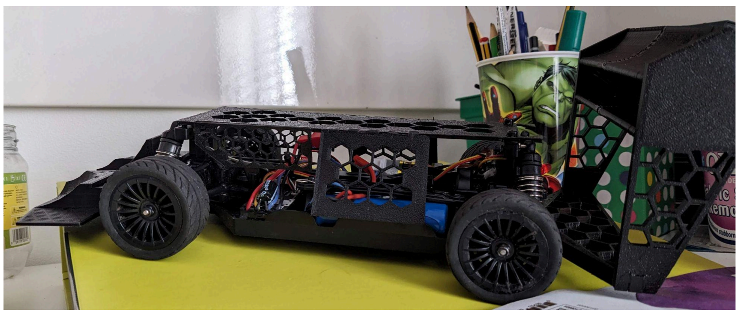


3.4 **Testing:** Testing each of my cars is a relatively unimportant part of this project. I can't set up a car, I can't make the changes for the next generation based on driver feedback. Whether that's due to my inexperience as a driver or the limitations of a £60 rc car I will leave the reader to decide… Nonetheless, I can feel that the grip increases each generation, considering now that, at full speed, I can turn and the car won't spin out. This was not the case with the stock bodywork. Despite this they are all fragile and susceptible to the wet. I believe that I have crashed every generation of my car. But this generation was the only one I didn't crash. I had the perfect opportunity from my dad to drive this car in a sports hall. In a sportshall I have all the grip I need, and a large space without curbs. As you can see to the left however, the car still ended up broken. Not from me, but the one time I let someone else drive the car.

4.0 **V3**

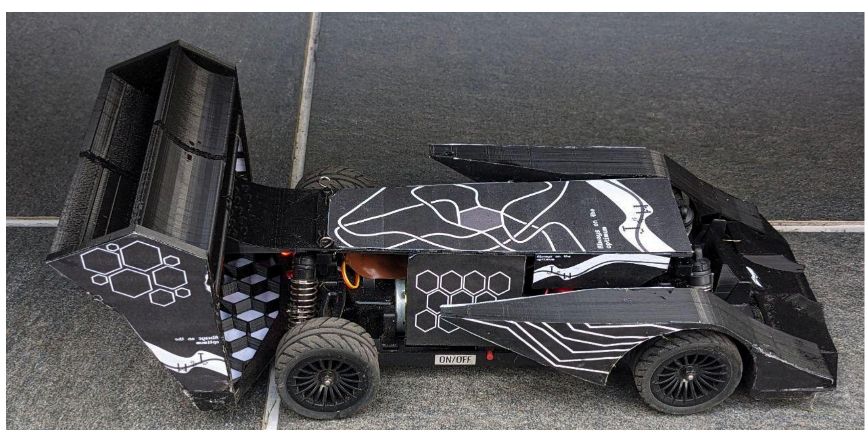
4.1 **Lightweight Design:** V3 was the first time I designed all the parts of my car to be crazy lightweight. As a comparison the V3 rear wing was just 6 grams lighter compared with the first two wings. But bear in mind that as I said previously, the wing had gained a significant amount in surface area. To accomplish these lightweight parts I used a combination of hex pattern lined with paper, spaces that weren't structurally important were completely





shelled out in CAD, or thin lines were added reinforcing parts in the direction of stress. I found that these lines are the best way to create lightweight parts, particularly in brittle PLA. They allow flex in one direction to absorb impact but are rigid in another to keep structures in their correct places, or transfer the minimal aerodynamic load I hoped I was generating.

4.2 **V3:** V3 was not an iteration in the aerodynamic design. But it was a huge step in the aforementioned lightweight design. As well as this the roof was completely different. It was fully 3D printed, and I remember spending an enormous amount of time adding finger joints and being sure that this would fit before printing. I remember that this car was tested very minimally. I crashed it whilst driving with my friend in Chichester and completely totaled the car. I think that doing that helped me. I looked at where the car was broken, and to be honest, was kind of impressed that crashing at full speed into a real van's tire hadn't broken the chassis at all. I was soon working on the next version of the car. It's also notable to state that I spent way too long designing the livery of this car. To be honest this was the only car I did this for, and I am glad of this. This isn't an art project after all.

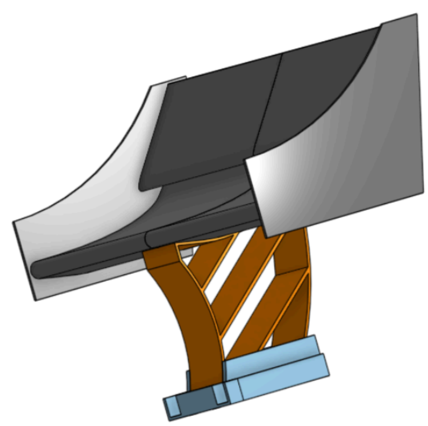


5.0 **AERORC EVO**

5.1 **An upgrade in everything…**

This car is the last I tested. It performed so well. Nothing wobbled. Nothing broke, and it was the only car, so far, that I haven't totaled. In terms of technical advancements, well, it mostly comes in the wings and lightweight design.

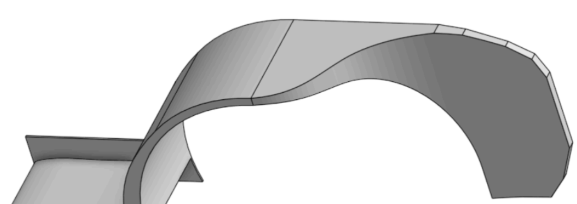
5.2 **Rear Wing**



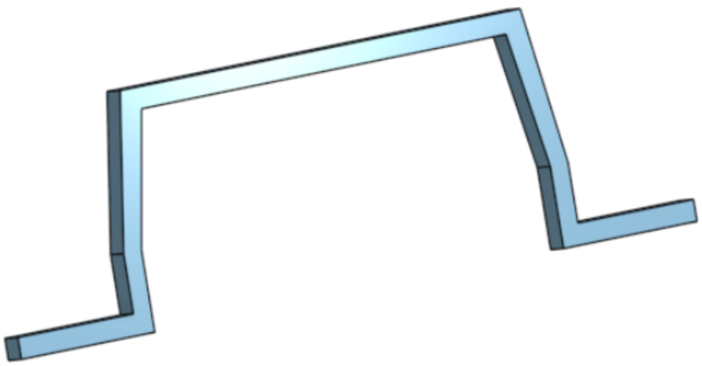
The rear wing on this car weighs just 38 grams. That's just over half the weight of the first wing I produced. And look at the size of it! Still though, It didn't use a real airfoil. As well as this, after all of the simulations for UDIPrototype, I now know just how ineffective this harsh second element would have been. Nonetheless this was an impressive iteration. In retrospect as well, the reason why this wing was so rigid was probably because of its low weight.

5.3 **Front Wing**

The front wing mounting was inspired by the F1 cars of 2000-2013, although there was no gap to the underfloor, it was mainly just to reduce the weight of



the mounting surfaces. It would have also acted as a splitter considering the high pressure that would have formed in this region. The front wheel cover design was greatly inspired by the Enviate Hypercar from Pikes Peak. To properly evacuate the wheel wake slightly smaller “fins” probably should have been used. Design for cars with radiators is something I am still developing. There is no “excuse” for adding out washing bodywork further inward as it would provide not much more than extra weight.



5.4 **Body cover**

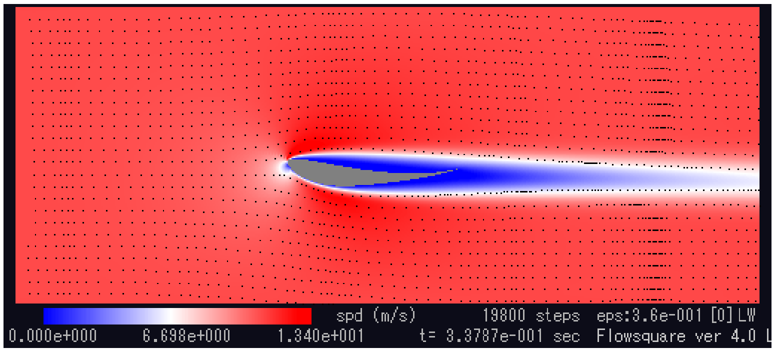
The bodywork is still something I am quite happy with. It is the epitome of lightweight design despite being fairly ugly. Essentially it had two 3D printed ribs with the actual cardboard cover glued in between. This made a lightweight but surprisingly stiff part. Sadly it did dissolve a little in the permanently wet car parks of England.

6.0 **Computational Fluid Dynamics**

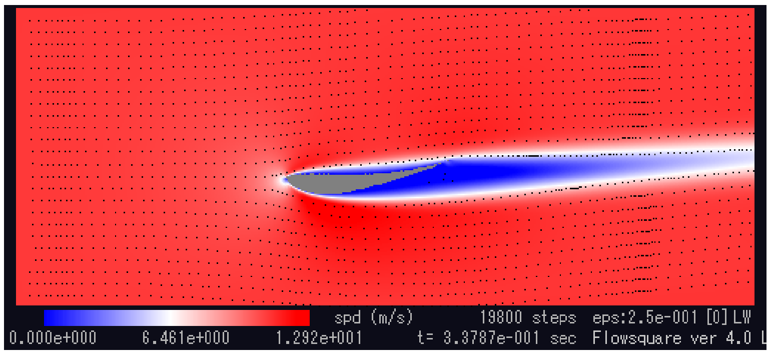
6.1: What is CFD? CFD is essentially a piece of software (just like Microsoft Excel or Chrome browser) which uses some complex equations to repeatedly find how air particles interact with an object. In my case that was the rear wing of the car. By finding this out I can analyse the flow around my rear wing and modify it to try and make it perform better in the next simulation. Somehow, this car’s rear wing has gone through ~104 hours of CFD. The software I use to perform these simulations is called Flowsquare. It can only simulate the flow around a 2D object. That is why I choose to only analyse and enhance the flow around my rear wing, which is just an extrusion of a 2D airfoil.

6.2: My method. I use a simple but repetitive method to enhance my Airfoil. I would run a simulation, change something, run the next simulation. For reference, each simulation takes around 8 hours to run on my slow PC. For the first 8 or so simulations my main focus was in understanding what kind of flow to expect around my stock selig 1223 airfoil, at the ultra low speed of 11 M/s.

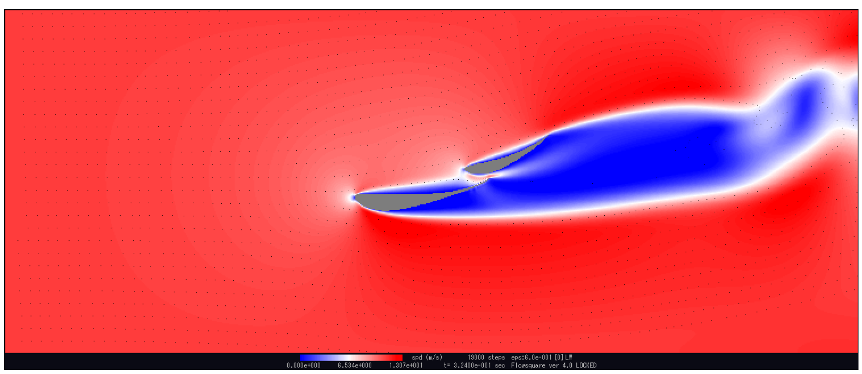
6.3: The first 7 simulations.



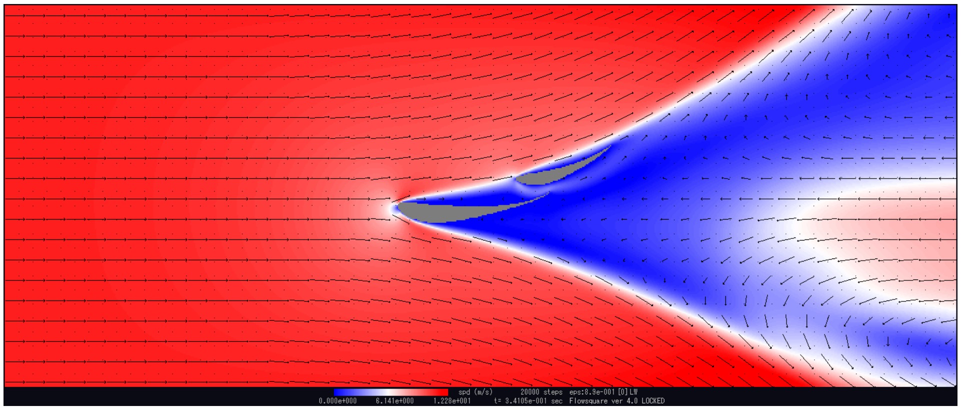
The stock Selig 1223 at a 0° angle of attack. Clear flow separation is visible but turbulence is minimale. It also hosts a healthy low pressure region.



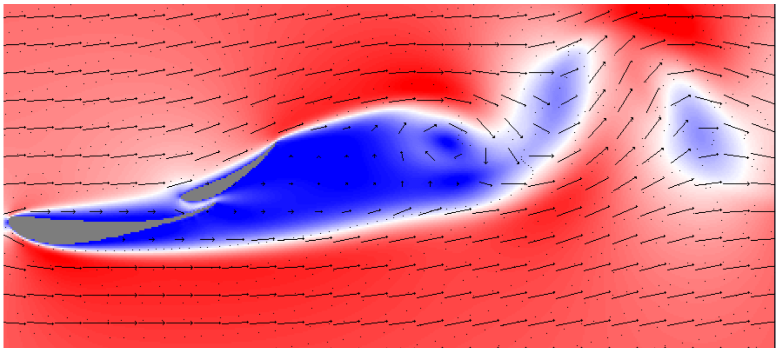
The stock Selig 1223 with enough angle of attack to make a flat section on the top of the wing. Has flow separation once again, as do all the airfoils, so I shall ignore this from now on. We can see that there is some “kick up” of air particles both under and over the wing.



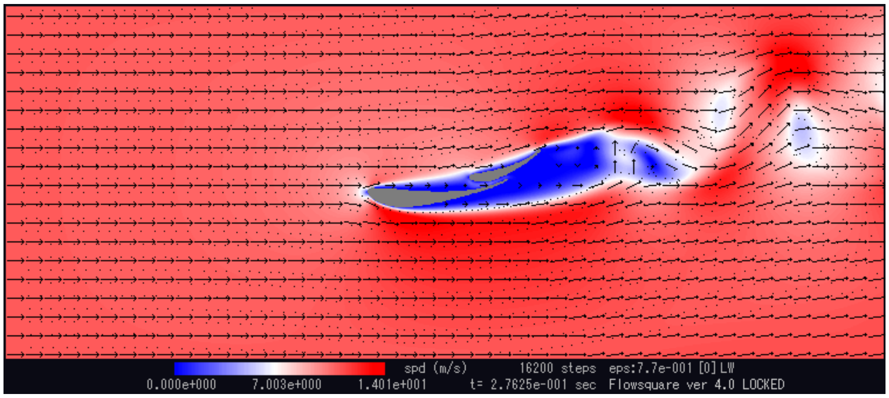
A lightly modified S1223, just modifying the top to create a nice flat surface. Also my first attempt at a second element. I went big knowing that things at this scale do very little. We can see a dramatic improvement in the upward movement of air, but also large drag inducing vortices forming. They are quite far from the trailing edge though.



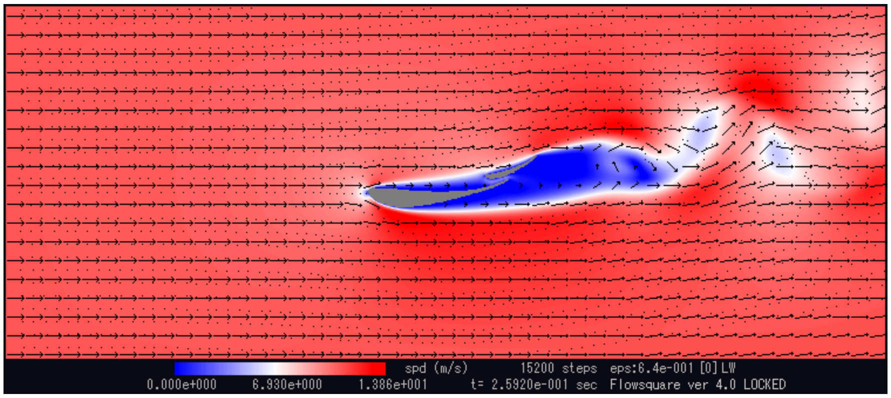
The Fourth simulation is a great example of my inexperience in CFD. It's clear to see this would not be the resultant flow from any wing in real life.



This simulation is one of the three where I enabled velocity vectors to visualise the flow in regions where vortices occurred. It's clear to see here just how turbulent the wake of these early simulations was.



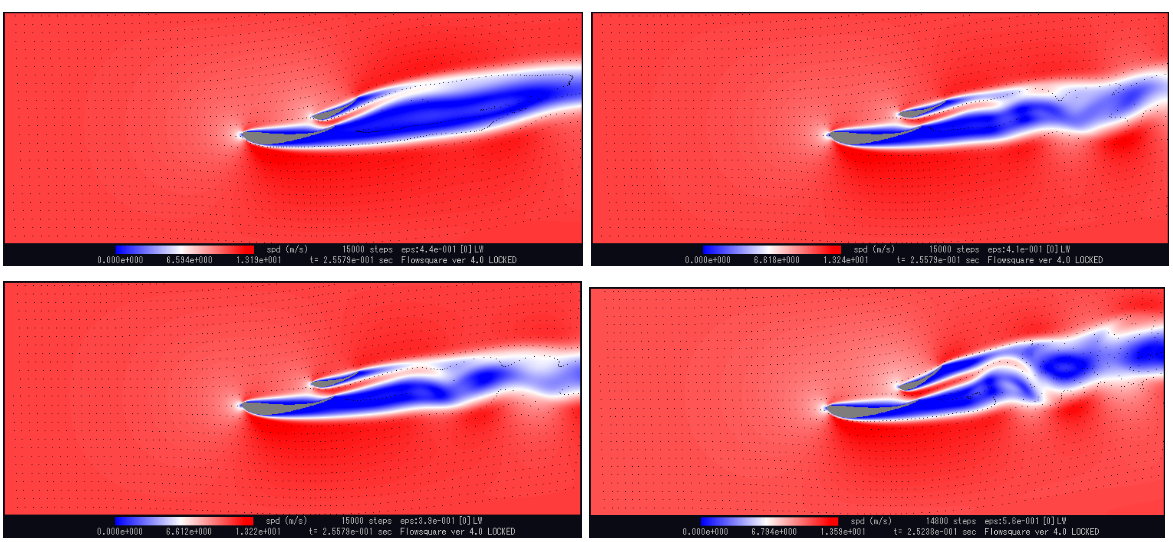
This iteration included a lower angle of attack with a second element very low to the first, and rather thin. In retrospect this was pointless, knowing that the second element should have a slot gap large enough to energise the immediate wake, which isn't happening. It essentially acts as one large wing here.



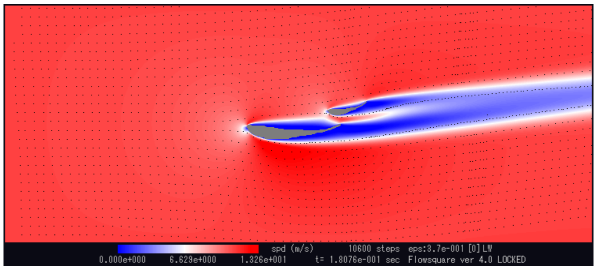
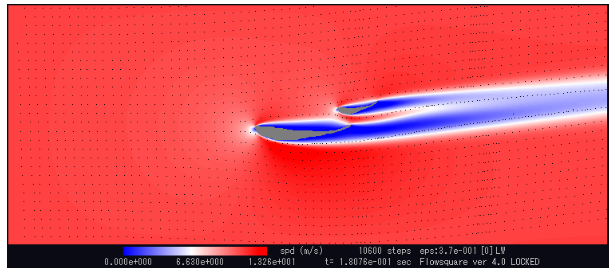
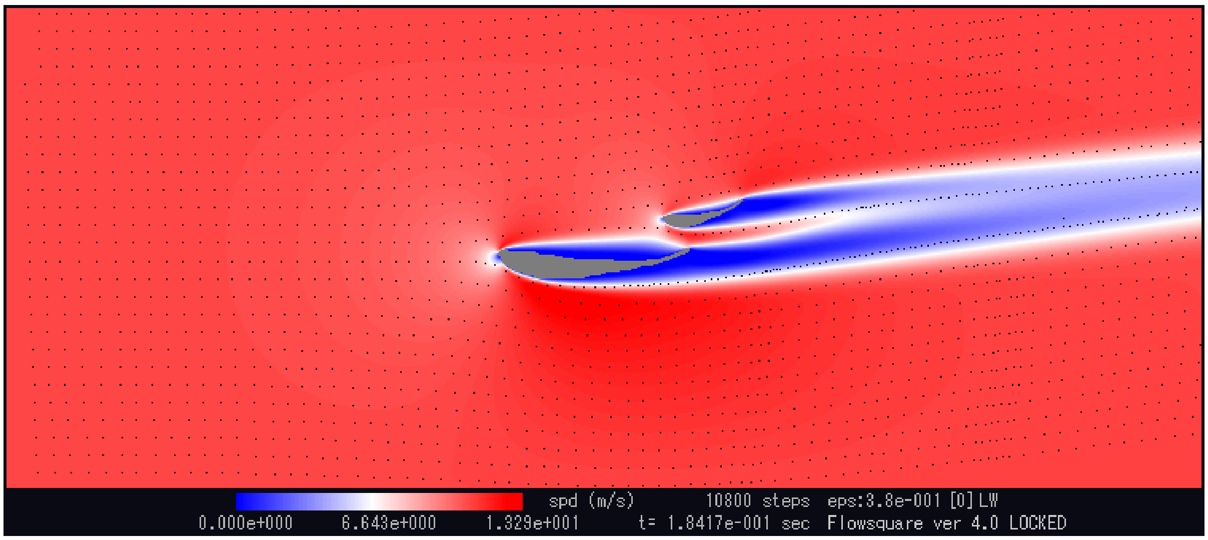
The final simulation didn't change the slot gap from the previous but did shorten the second element significantly. I am not sure if I noticed at the time but running these low angle of attack 1st elements isn't so effective at reducing drag when the flow separates at the leading edge.

6.4: The rest of the simulations.

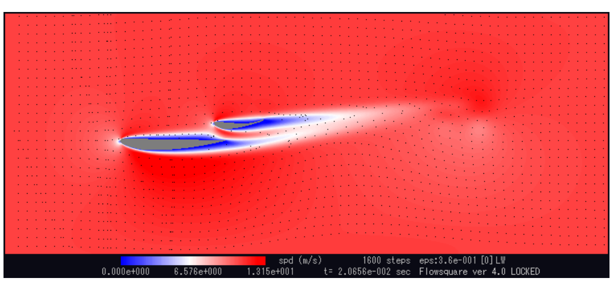
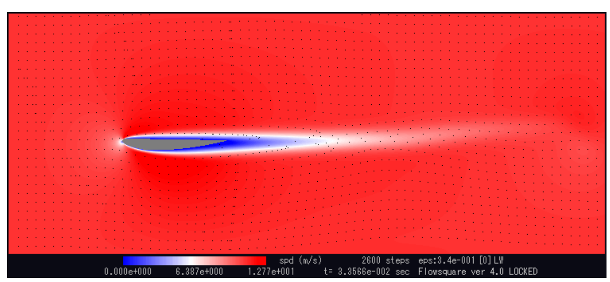
After the first eight preliminary simulations there was a clear need for more simulation power. Since each of these simulations was eight hours stand alone I could only run one or two per day, when my ideas for the next simulations were greatly exceeding that. To combat this I changed the cycle of development greatly, from just one at a time to up to four. The reason why I hadn't done this previously is because I didn't believe my PC had enough power. In reality it could run these four simulations simultaneously, the only thing changing being the loudness of the fan.



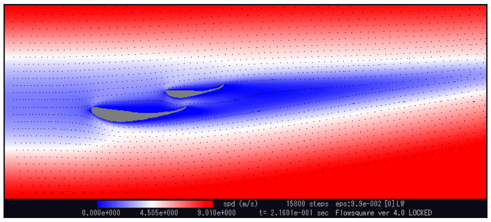
Varying in the second element angle of attack. 2,3,4 all create a turbulent wake whereas 1 has a solid low pressure zone. This makes 1 the best here. We can also see that the slot larger slot gaps in all of these airfoils allows for the low pressure zone behind the second element to be energised somewhat.

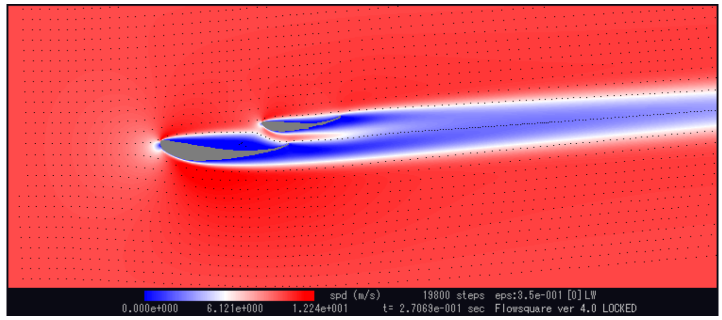




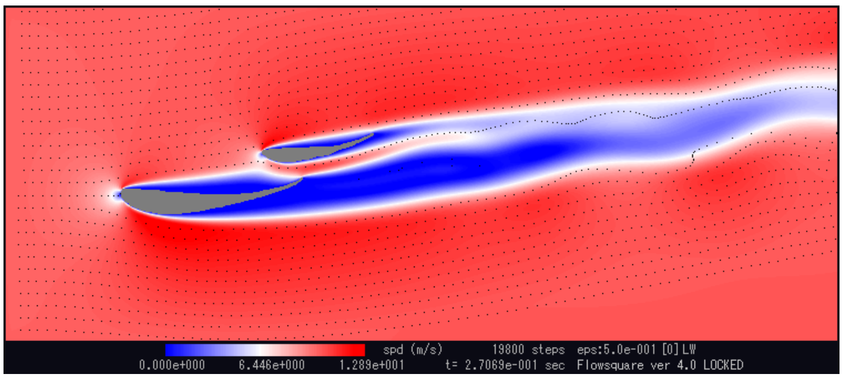
Tiny variations in angle of attack of a small second element. This doesn''t really do anything between simulations.

At this point I started to question whether getting the flow to attach was possible at all. To test this out I tested the Selig s7055 which has a flat top. We can see quite clearly in this example that the wake (in any dual element design at this speed) is somewhat energised and produces more kick up in the flow behind the wing. This was one of the more important simulations as it allowed me to ground my expectations of an attached flow airfoil. From there I could optimize for as high downforce as possible **with flow separation**.



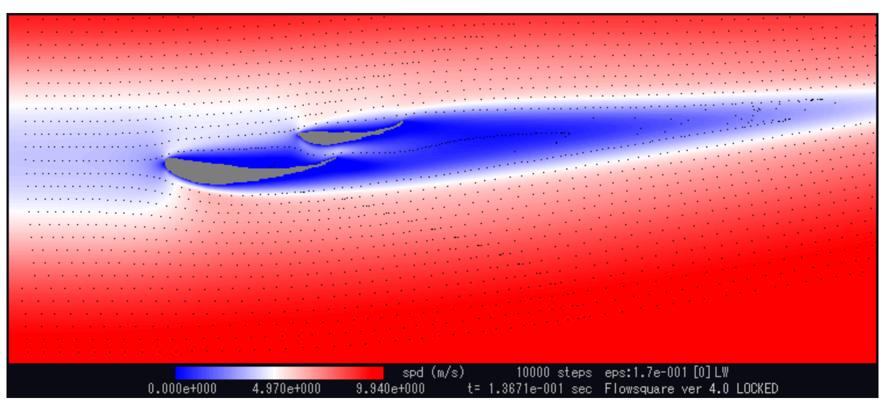




Here was another round where only small variations were made in the angle of the second element. However we can see that this tiny variation of 5° results in a much more turbulent wake in the second than the first, but also a lot more air moving vertically, resulting in more down force. We can also see what happens when the wing is placed too close to the front of the boundary. The low pressure zone propagates to the front of the boundary ruining the simulation.

Round 6 was relatively meaningless so I won't include pictures. It was another small variation simulation, which was changes of +-2° on the non-turbulent simulation 2 from the second round. In retrospect this was the wrong direction, considering just how much more air is moved vertically when a turbulent wake is permitted.

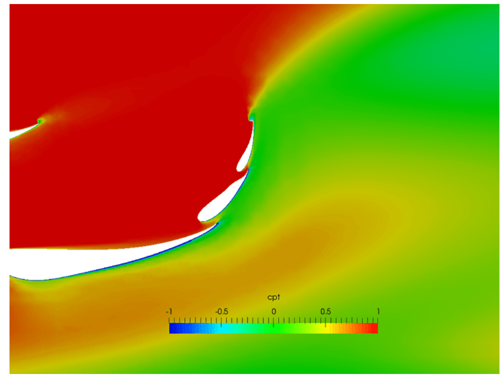
Luckily, in **round 7**, I pressed forward with the +2° version of round 6. This resulted in a less turbulent wake than that from round 5, but nearly the same amount of vertical air movement. As well as this I varied the slot gap to optimise this final variable. (Note the propagation is happening again, it doesn't effect the flow so much though, just the speed of incoming air in general)



This was the final simulation, and on looking back and analysing these I am quite content with the results I achieved. Nonetheless there are improvements and changes I would make If I did this again.

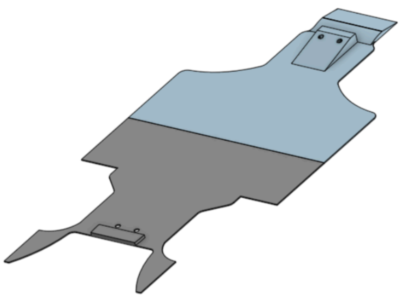
6.5: What I would do now.

1.) Try a 3rd element.
2.) Rotate the entire dual / triple element assembly.
3.) Embrace the low pressure zone (optimize for as large a low pressure zone as possible without large vortices.)



We can see to the right a Formula SAE rear wing, which is really the only equivalent to a wing optimised at low speeds like mine. All of the things addressed above are seen here, so the direction of development would be clear. Note that although it runs at low speeds it also is much larger. I haven't neglected to realise this…

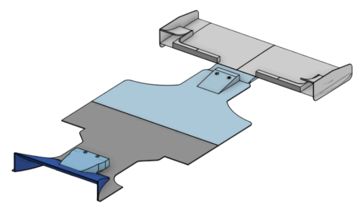
7.0 **A different approach: UDIPrototype.**

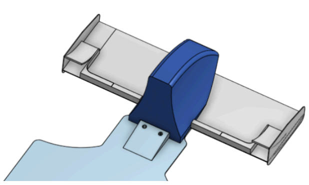


Instead of designing a front, a rear and a body in the middle, I instead decided to try and design the car as one. The front is mounted with two screws like the previous version, as is the rear. The difference being that I use an extremely thin floor connected to both of these sides as a mounting point.

I'll show over the next few pictures how the car builds up from here.

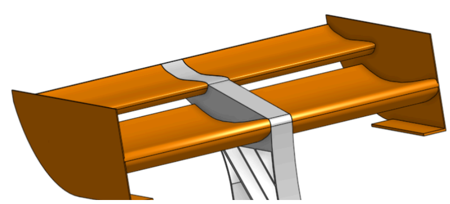
Firstly the large flat top front wing (Selig s7055 airfoil) is glued to the front plate. The diffuser attacher is mounted with the stock screws through the rear plate and into the stock chassis. The diffuser is glued to this attachment and the rear plate as well, creating one rigid structure. Seen to the right.

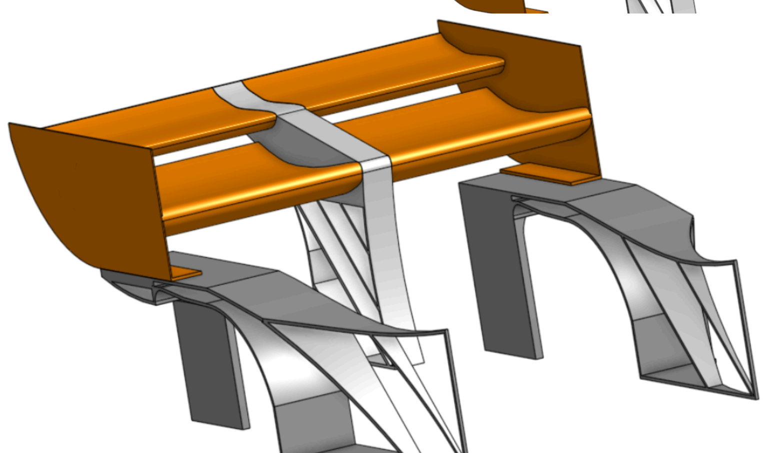




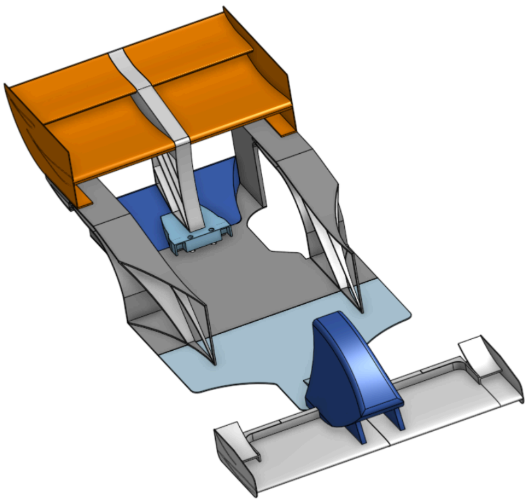
The hollow front nose cone is then glued onto the wing and the front base plate, creating a strong 3D printed sandwich.
Seen to the left.

There is a sort of sub assembly in the top of the rear. The two sections of the wing are glued to the wing spine (which could have been thinner in retrospect).
Seen to the right



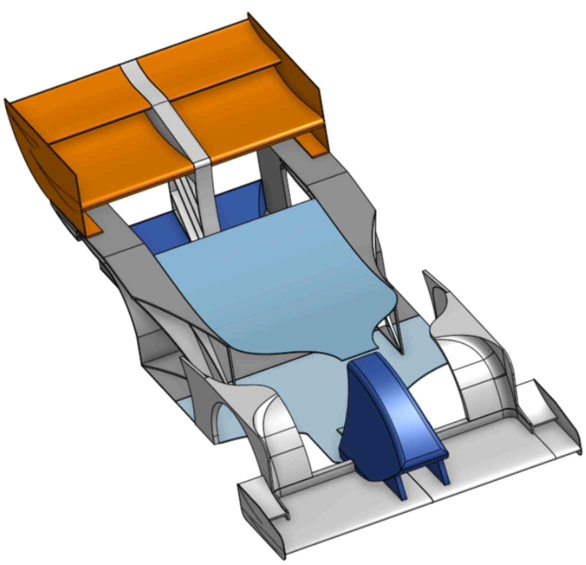


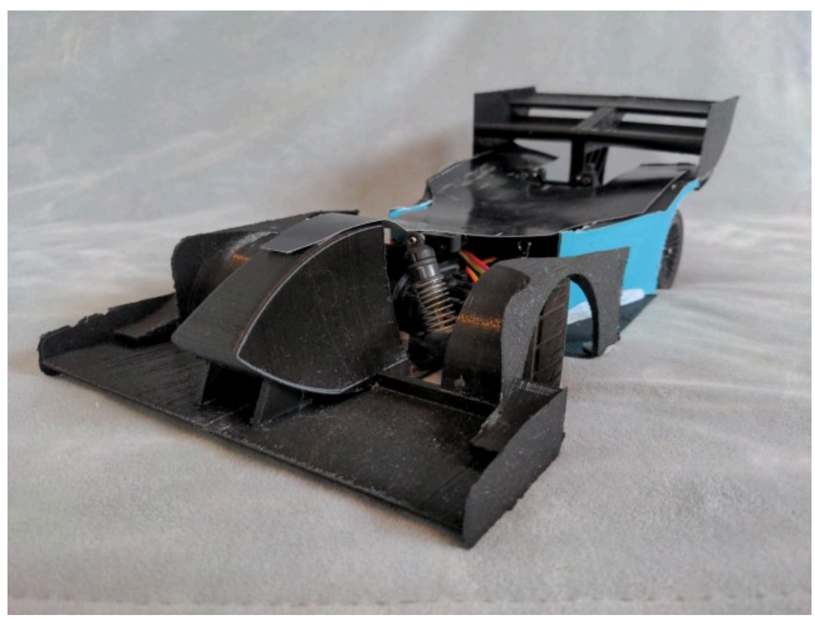
The side of the body is glued to the bottom of the wings (the simulated and modified S1223). This forms a substantial part of the rear of the car. You can see here a lot of that all important lightweight design I have learnt. The large sections of the hollow space are filled only with a few braces. These allow some flex to occur but not nearly as much if these parts were truly hollow, whilst only adding a few grams. These pieces are then wrapped in either paper or card. This makes them look much less professional, but are much more effective than solely 3D printed from a lightweight and performance perspective. Seen above.

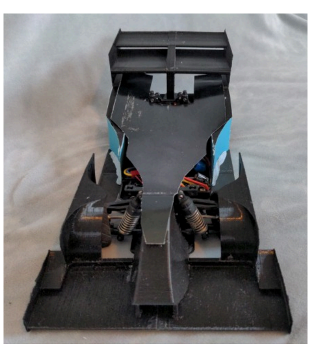


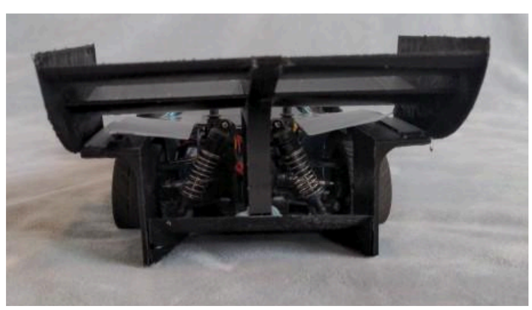
This rear wing assembly is attached only with double sided sticky tape and blue tack. This is not because I don't know how to use glue, but because I needed these parts to be removable, and I didn't have nuts and bolts yet.. Seen left.

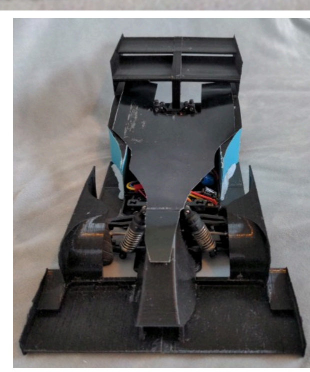
Finally the front wheel covers are glued to the front wing and blue tacked to the undertray. This is again in the spirit of removability. And the car is complete. We can see now that everything is connected to everything. It creates one strong, rigid and lightweight part. This aids vehicle dynamics, and the cars ability to accept aerodynamic load. Seen below.











Final Remarks and The Future.

This project has been a whole year of personal improvement. I have learnt CAD to a *decent* level, learnt the basics of aerodynamics and CFD, learnt how to engineer lightweight parts at the RC car scale, and become enormously invested in the world of motorsport. But most importantly I have had so much fun. From crashing the car and rebuilding it to finishing UDIPrototype, I have always had fun. At no point have I felt bored of this project, and I could quite happily continue to develop this car. But I know I have so much more in me than that. So, the future:

I have been considering the idea of a fully custom concept/hyper car for a while now. Like Indeterminate Designs' RC Hypercar. But I have always been non-commital to this. I started designing a car for the Mantium Challenge (a virtual F1 design championship) and realised that was doable for me with my current knowledge of CAD. I started my own 2D version of the Mantium Challenge. But it's not the same as the buzz I get from seeing my designs in real life. That's why I am commiting to the Hypercar project. It's not just going to teach me aerodynamics. It's going to teach me **everything** about how to design a racecar. And that's going to be hard...

Will Kennedy, 2024.

Note that at the time of writing I haven't actually tested UDIPrototype, which is why I didn't mention its performance previously. I will add remarks here after testing it. I wanted to finish and cut ties with this project before 2025, which is why I haven't waited.